NUS-ISS

Problem Solving Using Pattern Recognition

Deep learning: Create

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How deep learning is used to create: Use cases

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Generative AI: The "Create" Category

The landscape has matured significantly. We now categorize models by the specific modality of content they generate from natural language instructions.

Text & Code

- Gemini 3.1 Pro (Google)
- GPT-5 / ChatGPT
- Claude 5 / Claude Code

Image Gen

- Nano Banana 2 (Google)
- FLUX.2 [pro]
- Midjourney v6+ / DALL-E 3

Video Gen

- Kling 3.5 (Kuaishou)
- Wan 2.6 (Alibaba)
- Veo 3 (Google)

Audio & Music

- MusicFX / MusicGen
- Suno / Udio
- ElevenLabs (Voice)

3D Objects

- Get3D (NVIDIA)
- Sloyd / 3DFY
- Luma Genie

Specialized

- Adobe Firefly (Commercial)
- FashionAI / DeepFashion
- GauGAN / DeepArt

Content Generation AI

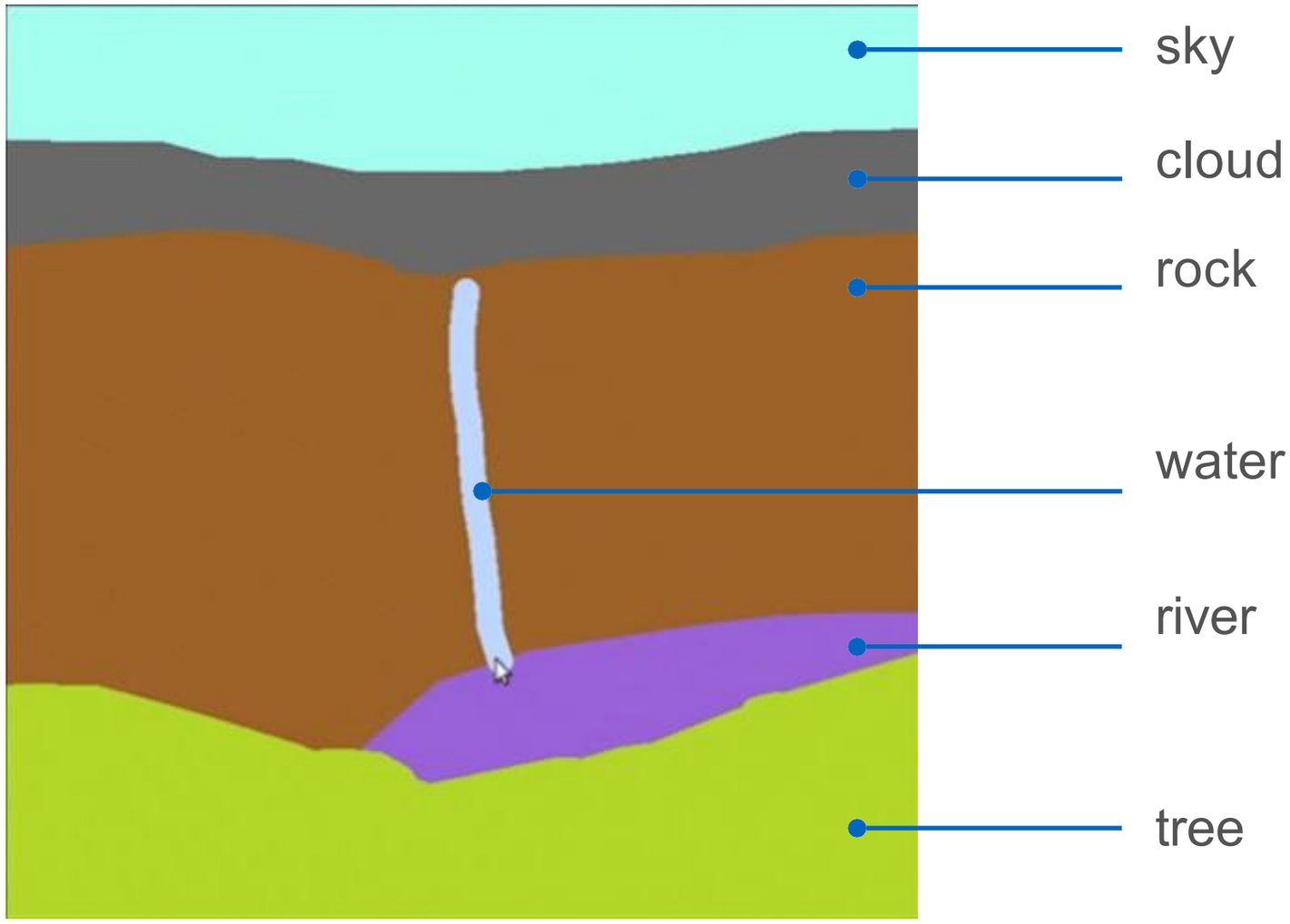
ChatGPT, Copilot, Gemini and Meta AI



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Drawing

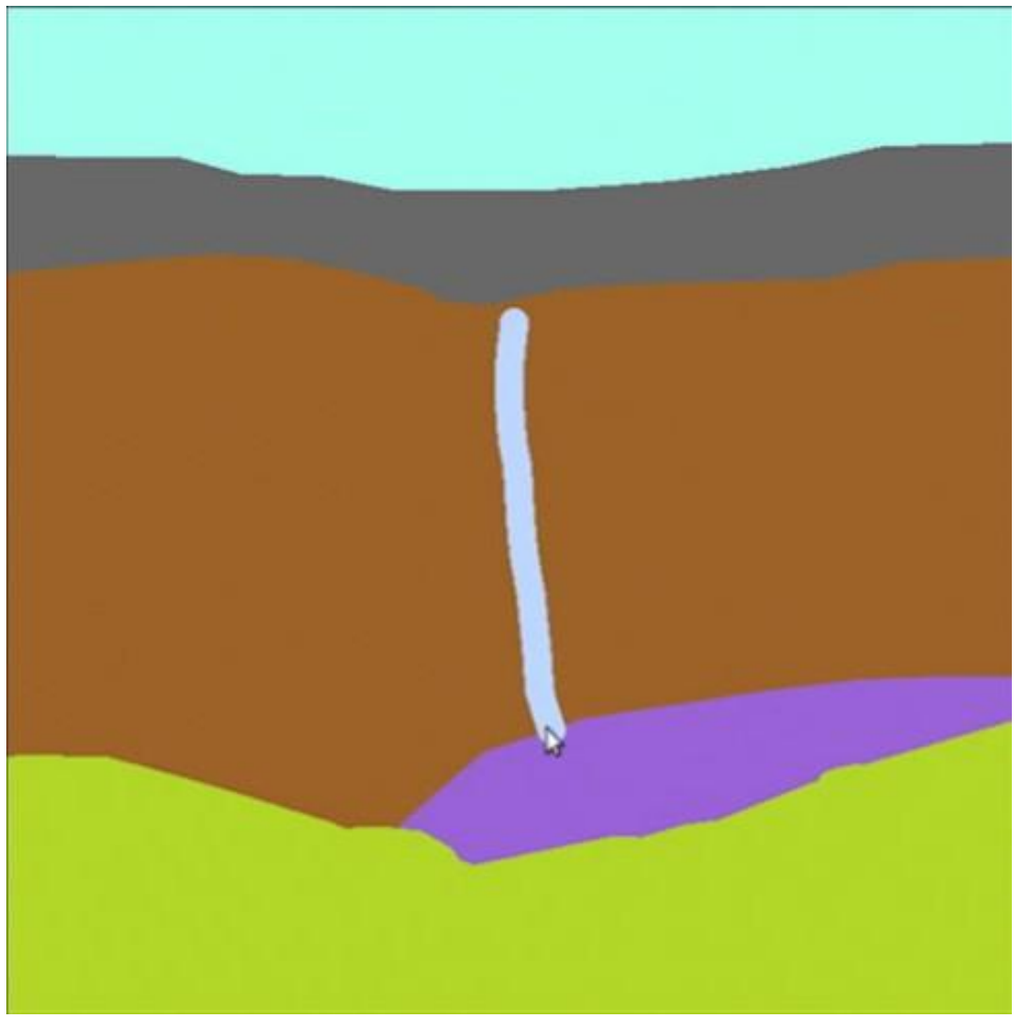
A sketch



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Drawing

A sketch





Source: https://blogs.nvidia.com/blog/2019/03/18/gaugan-photorealistic-landscapes-nvidia-research/

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Drawing

GauGAN (inspired by Gauguin)

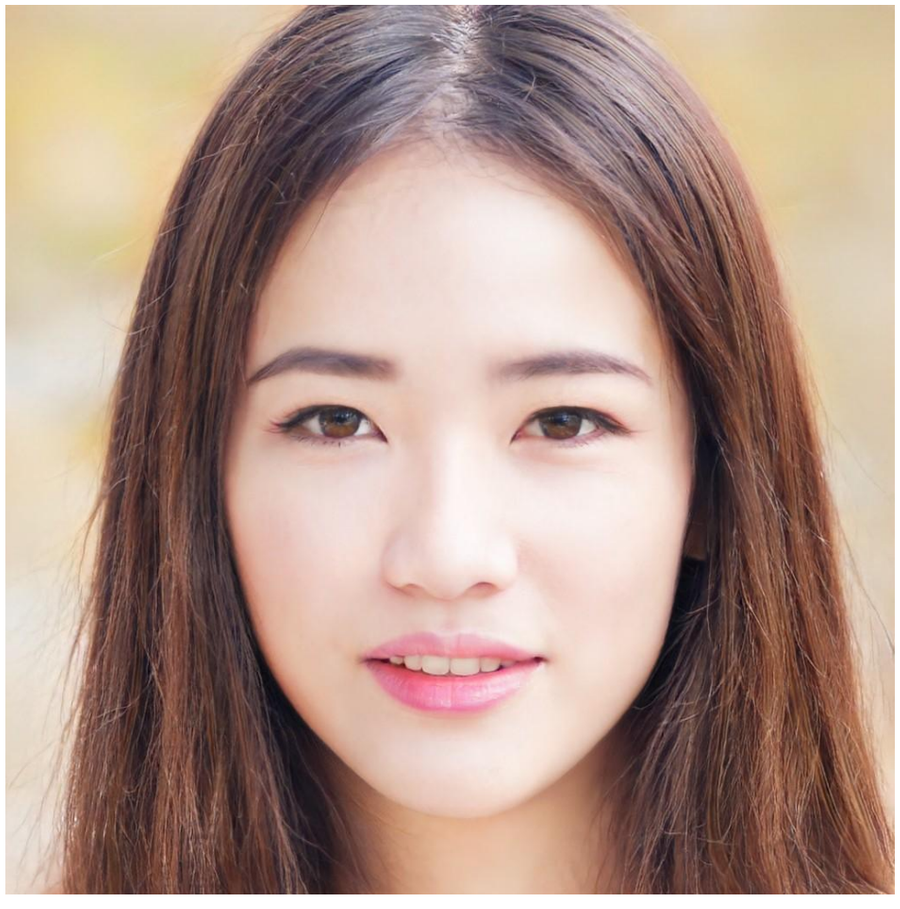


Source: https://blogs.nvidia.com/blog/2019/03/18/gaugan- photorealistic-landscapes-nvidia-research/
https://www.youtube.com/watch?v=p5U4NgVGAwg

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Do you know...

Who she or he is?





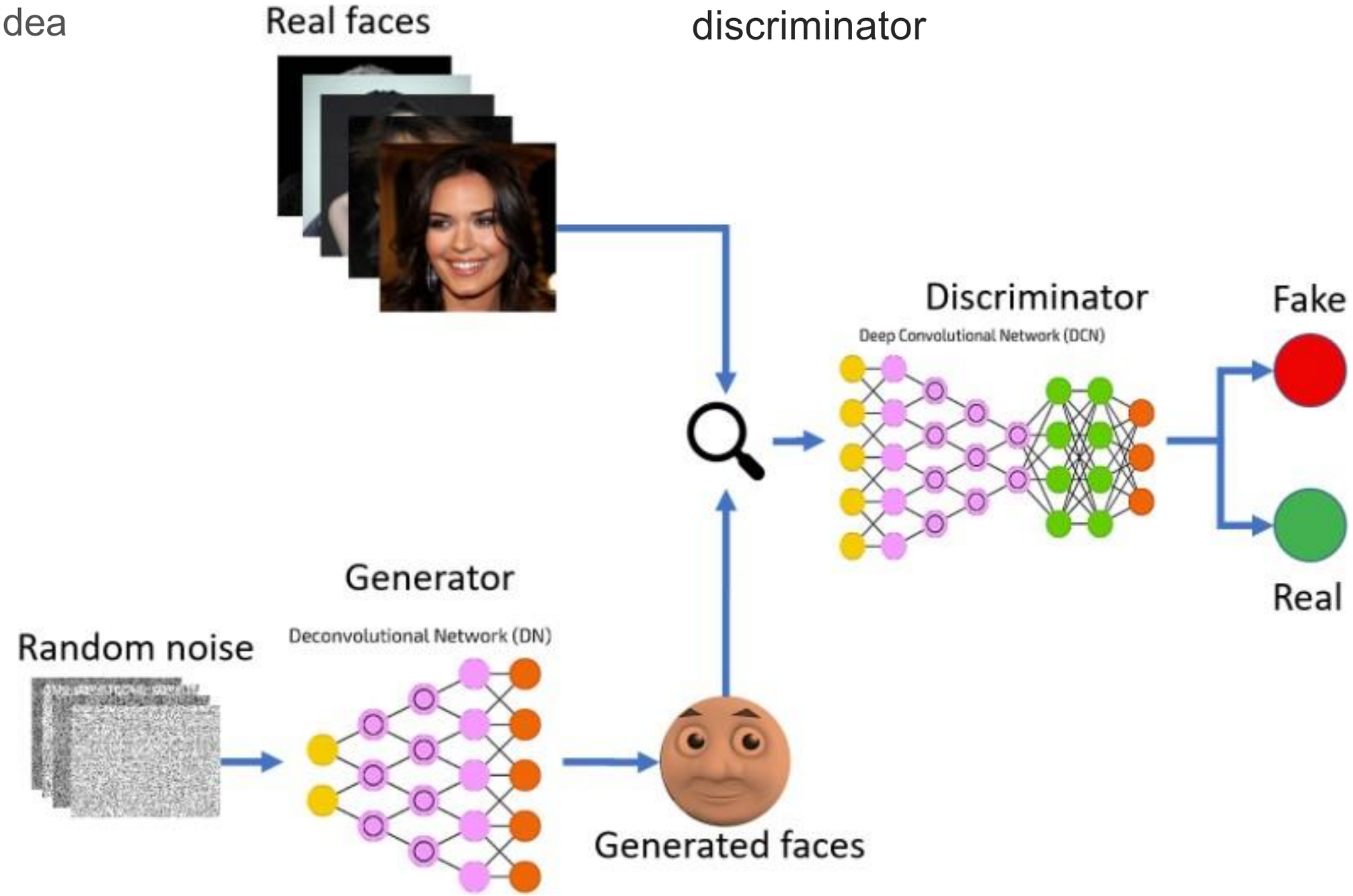
Source: https://thispersondoesnotexist.com

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Generative Adversarial Networks

The idea

- The generated faces and the real faces take turn to be fed into the discriminator

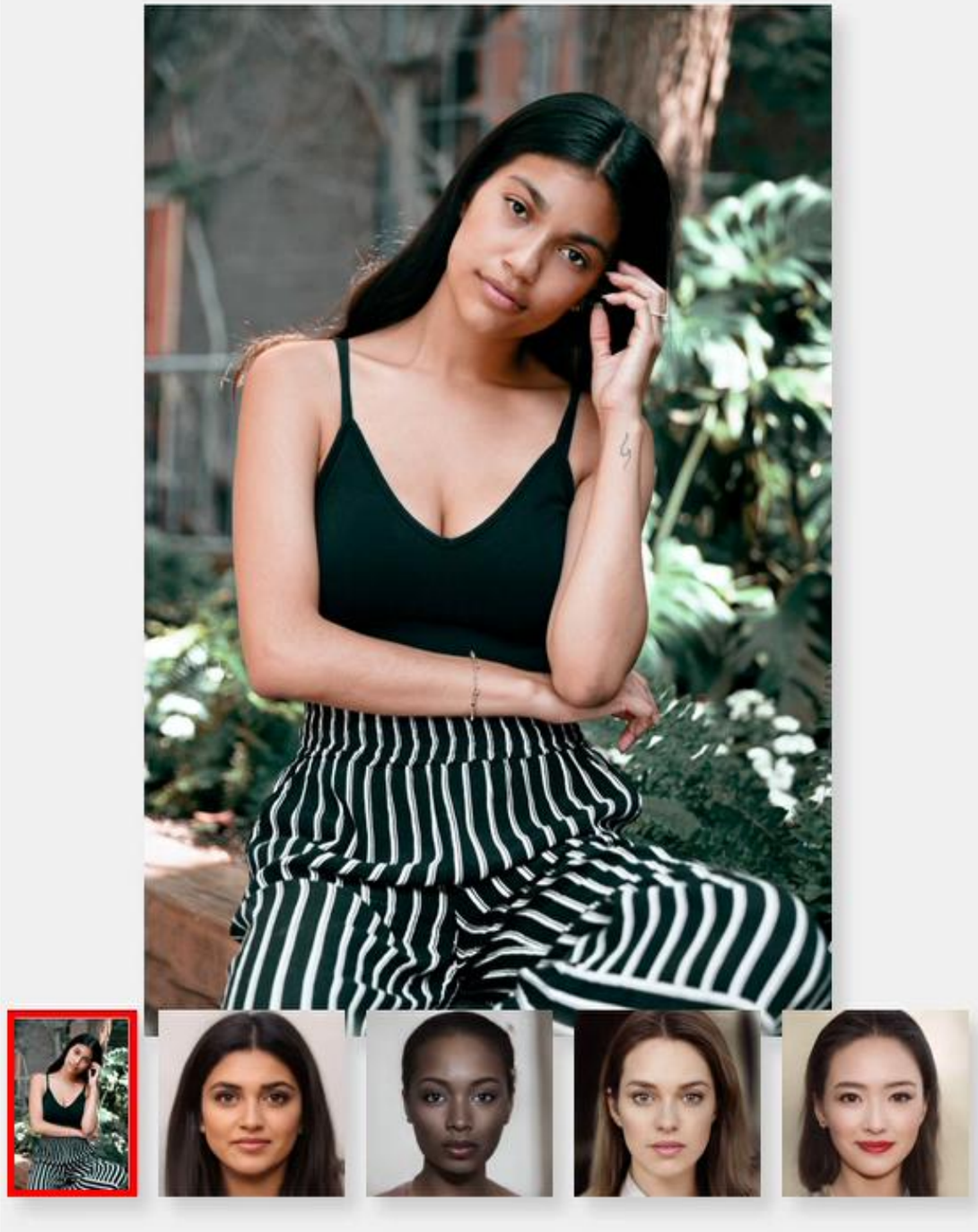


Source: https://www.spindox.it/en/blog/generative-adversarial-neural-networks/

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Change face

Unlimited possibilities



Source: https://app.generative.photos

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Practical usage
fashion



Which ones contain the actual model?

Source: https://qz.com/1090267/artificial-intelligence-can-now-show-you-how-those-pants-will-fit/

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Generative Adversarial Networks

styleGAN



Source: https://www.youtube.com/watch?v=q4U0VDrd60w

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Generative Adversarial Networks

Deepfake



Source: https://www.youtube.com/watch?v=S1ba8etKGBY

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Generative Adversarial Networks

Deepfake



Source: https://www.youtube.com/watch?v=F4G6GNFz0O8

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HeyGen: AI-powered video creation tool

Translate Video into ANY Language with Lips Correctly Synced

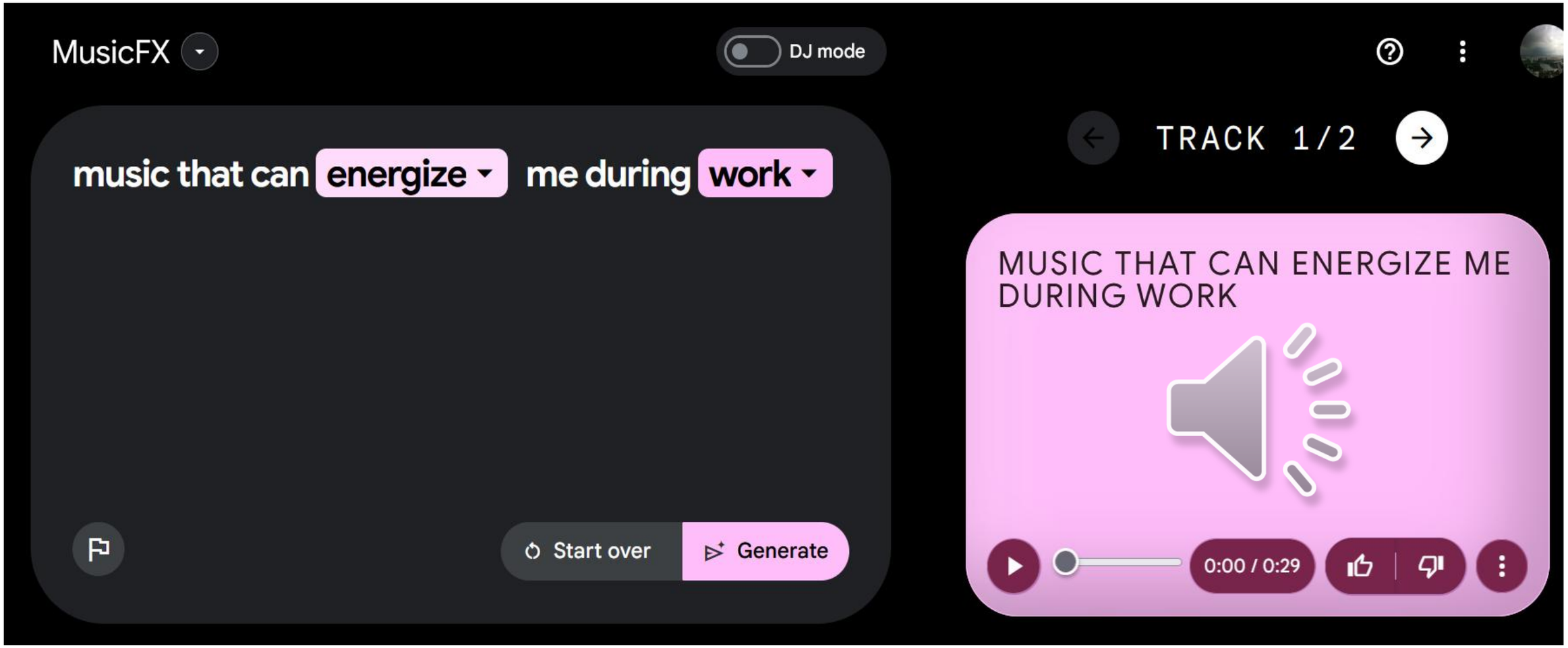


Source: https://www.heygen.com/;
https://www.youtube.com/watch?v=h5JNnvXjmmY&t=74s

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MusicFX

Music Generation based on Text Prompts

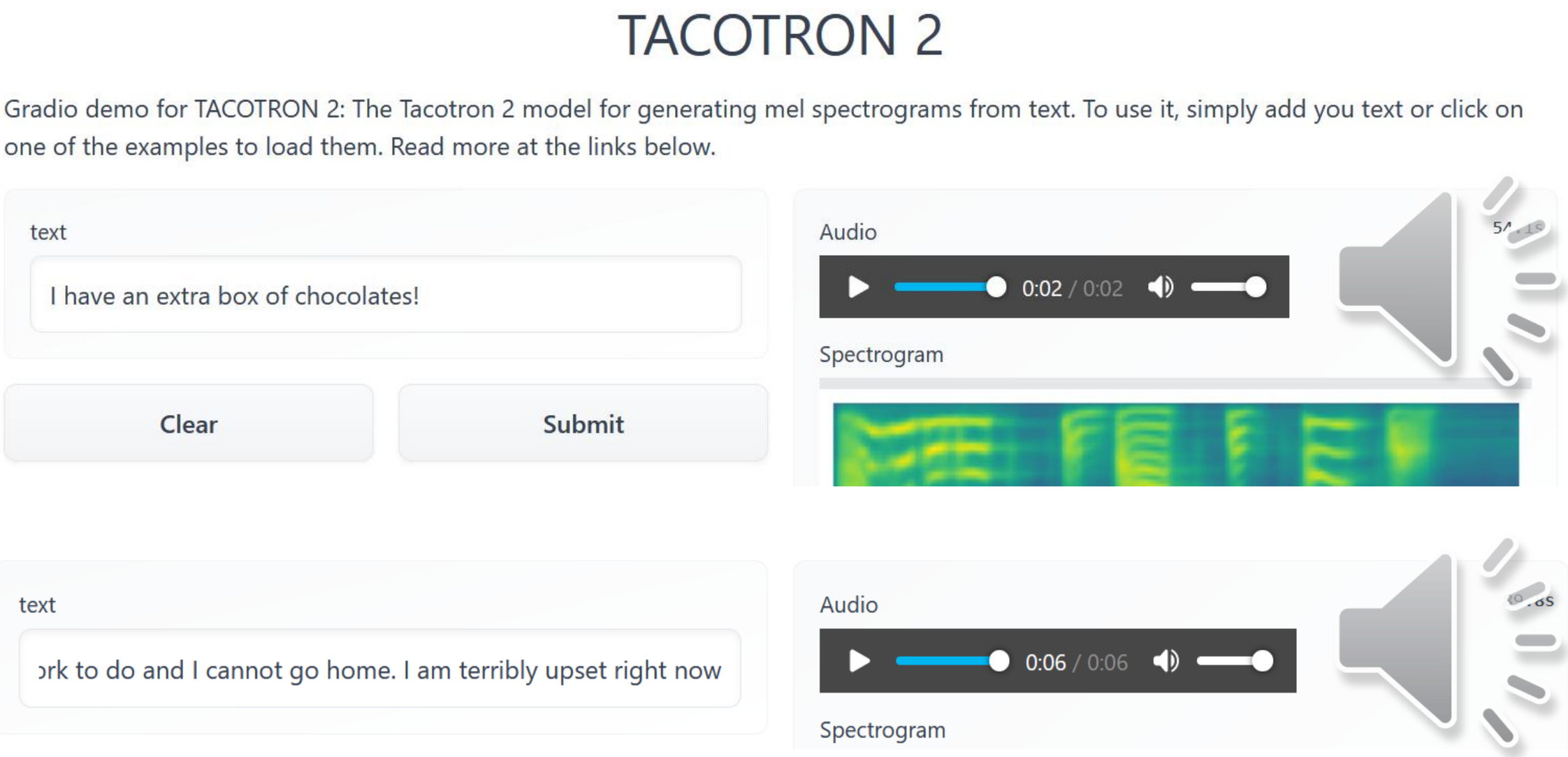


Source: https://aitestkitchen.withgoogle.com/tools/music-fx

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TACOTRON 2

Speech Generation



Source: https://huggingface.co/spaces/pytorch/Tacotron2

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Disappearing people

The guy is missing ….



Source: https://github.com/jasonmayes/Real-Time-Person-Removal

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Generative Adversarial Networks

How about ... anime?



Source: https://crypko.ai

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Generative Adversarial Networks

How about ... artpiece?

- Sold through auction at Sotheby



Source: https://news.artnet.com/market/artificial-intelligence-sothebys-1481590

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How about 3D in a scene?

Generates from images



Source: https://github.com/facebookresearch/synsin

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